

Instructions for Phase 1

The first part of the exercise will be done on CP2K lab.

The second and third part will be done on the **aiidalab** (mmm-course.aiidalab.io)

Pull the exercise from the GIT from the **mmm-course.aiidalab.io** as usual:

```
git clone https://github.com/cpignedoli/MMM-2026.git
```

- Go to the **Drill_Passerone** directory
- There are three notebooks: Download the **CP2K_Lab_slabs.ipynb** on your local laptop
- Open [cp2k Lab](#) (see Exercise 4 and 5, instructions on the moodle or from the exercises directory)
- Upload the **CP2K_Lab_slabs.ipynb** file to cp2k Lab

ON CP2K Lab

- Execute the cells to create the models to simulate, follow the instructions.
- Once all simulations are done, you can download the optimized structures (of the slab+molecules) to your laptops, and upload them to **mmm-course.aiidalab.io**

ON mmm-course.aiidalab.io

- (from the **mmm-course.aiidalab.io**) open the **aiidalab-installations.ipynb**, and install the special version of lammmps with reaxff support
- Open the **aiidalab-reaxff-slabs.ipynb**
- This can load all the structures and then you can compute adsorption energies.

CO Electroreduction on Cu(100): DFT Adsorption Energies and ReaxFF Metadynamics

Course	Master in Materials Science / Computational Chemistry
Tools	CP2K lab • ASE • LAMMPS • (with) PLUMED
Estimated time	Phase 1 (DFT): ~4 h Phase 2 (ReaxFF + MetaD): ~4 h
Reference paper	Calle-Vallejo & Koper, Angew. Chem. Int. Ed. 2013, 52, 7282–7285 (paper available in the Exercise dir, including SI)

1. Scientific Background

Copper is the only metal capable of electroreducing CO₂/CO beyond formate, producing hydrocarbons and oxygenates. The Cu(100) facet is particularly selective towards C₂ products and forms them at low overpotentials without significant C₁ by-products.

Calle-Vallejo & Koper (2013) propose that the rate-determining step is the C–C coupling:



The electron transfer and proton transfer are decoupled: the electron transfers first (forming the negatively charged dimer), then the proton arrives. This explains why C₂H₄ formation is pH-independent on the SHE scale, unlike CH₄.

In this exercise you will compute the three key DFT energies that characterise this step, then calculate the C–C coupling barrier using ReaxFF metadynamics and connect the two via the Computational Hydrogen Electrode (CHE) formalism.

2. Learning Objectives

1. Set up and run DFT geometry optimisations with CP2K for a periodic Cu(100) slab.
2. Compute adsorption energies of *CO, 2*CO, and *C₂O₂ (neutral) and understand the CHE formalism.
3. Connect a neutral DFT calculation to the electrochemical scale using the electron affinity of the dimer.
4. Set up a well-tempered metadynamics simulation in LAMMPS + PLUMED along the C–C distance.
5. Reconstruct the free energy surface and extract the C–C coupling barrier.
6. Critically compare the neutral DFT result with the ReaxFF metadynamics barrier.

Phase 1 — DFT with CP2K: Three Groups of Calculations

All three calculations share the same slab (3×3×4 Cu(100), 15 Å vacuum, bottom 2 layers fixed) and the same CP2K settings. Run them all at neutral charge ($q = 0$, MULTIPLICITY 1). Collecting all three energies allows you to build the partial free energy diagram of Figure 3a of the reference paper.

Do all the calculations in a similar way as in the exercise “Exercise_05_Ethanol”

CP2K Settings (common to all three groups), computed on CP2K Lab

Parameter	Value
Functional	PBE + D3 dispersion (Grimme)
Basis set	DZVP-MOLOPT-GGA-GTH
Pseudopotential	GTH-PBE
Plane-wave cutoff	400 Ry (relative cutoff: 50 Ry)
SCF convergence	1×10^{-5} Hartree
Charge	0 (all calculations neutral)
Multiplicity	1 (closed shell)

Group 1 — Single *CO on Cu(100)

What to compute

Optimise the geometry of one CO molecule adsorbed on the top site of Cu(100). Also optimise the clean slab and an isolated CO molecule (in a box matching the slab cell). Collect:

- $E(\text{slab})$ — clean slab, same supercell
- $E(\text{CO}^{\text{Gas}})$ — isolated CO in the slab cell, Γ -point only
- $E(\text{slab}+\text{CO})$ — one CO adsorbed on top site

ASE snippet

```
from ase.build import fcc100
from ase.constraints import FixAtoms
from ase import Atoms
from ase.io import write

slab = fcc100('Cu', size=(3,3,4), vacuum=15.0)
slab.center(vacuum=15.0, axis=2)
fixed = [a.index for a in slab if a.position[2] < slab.cell[2][2]/2 - 5]
slab.set_constraint(FixAtoms(indices=fixed))
```

```
# Place CO on top site (C down, O up)
z_top = slab.positions[:,2].max()
co = Atoms('CO', positions=[[0,0,z_top+1.3],[0,0,z_top+2.45]])
co.translate([slab.cell[0][0]/2, slab.cell[1][1]/2, 0])
slab_co = slab + co
write('slab_1CO.xyz', slab_co)

# Isolated CO: same cell as slab
co_iso = Atoms('CO', positions=[[0,0,0],[0,0,1.15]])
co_iso.set_cell(slab.get_cell())
co_iso.center()
write('CO_isolated.xyz', co_iso)
```

Adsorption energy

$$\Delta E_{\text{ads}}(*\text{CO}) = E(\text{slab}+\text{CO}) - E(\text{slab}) - E(\text{CO}^{\text{GAS}})$$

Expected value after ZPE and entropy corrections: -0.25 eV (paper, Fig. 3a). The raw DFT value will be more negative (~ -0.5 to -0.7 eV); the difference is due to the large entropic cost of CO(g) at 298 K ($-TS \approx +0.3$ eV from Table S.1 of the SI).

Group 2 — Two *CO on Cu(100)

What to compute

Place two CO molecules on adjacent top sites of the $3 \times 3 \times 4$ slab, separated by one Cu-Cu nearest-neighbour distance ($a/\sqrt{2} \approx 2.57$ Å along the [110] direction). Optimise and collect $E(\text{slab}+2\text{CO})$.

ASE snippet

```
a_Cu = 3.634
nn = a_Cu / 2**0.5 # nearest-neighbour distance
z_top = slab.positions[:,2].max()

top_atoms = [a for a in slab if abs(a.position[2]-z_top) < 0.1]
top_sorted = sorted(top_atoms, key=lambda a:(a.position[0],a.position[1]))
anchor = top_sorted[4]
x0, y0 = anchor.position[:2]

CO1 = Atoms('CO', positions=[[x0, y0, z_top+1.3],
                             [x0, y0, z_top+2.45]])
CO2 = Atoms('CO', positions=[[x0+nn, y0+nn, z_top+1.3],
                             [x0+nn, y0+nn, z_top+2.45]])

slab_2co = slab + CO1 + CO2
write('slab_2CO.xyz', slab_2co)
```

Adsorption energy

$$\Delta E_{\text{ads}}(2*\text{CO}) = E(\text{slab}+2\text{CO}) - E(\text{slab}) - 2 \times E(\text{CO}^{\text{GAS}})$$

Expected: -0.40 eV (paper). The second CO adsorbs ≈ 0.15 eV less strongly than the first due to lateral repulsion between the two *CO adsorbates.

Group 3 — *C₂O₂ Dimer on Cu(100), Neutral

What to compute

Optimise the neutral *C₂O₂ dimer on Cu(100). Use the asymmetric top-bridge geometry from Fig. 3b of the paper as starting point (provided by the build script). Collect E(slab+C₂O₂).

Starting geometry (from build script)

The dimer has an asymmetric configuration:

Atom	Role	Height above surface	Bader charge (paper)
C ₁	Top site, bonded to surface Cu	+1.30 Å (start)	2.03 e ⁻
O ₁	Bridge contact, bent toward adjacent Cu	+0.60 Å (start)	7.48 e ⁻
C ₂	Upper arm, no direct surface contact	+2.42 Å (start)	3.52 e ⁻
O ₂	Terminal, points toward vacuum	+3.32 Å (start)	7.91 e ⁻

The Bader charges sum to ≈ 21 e⁻ (vs. 20 e⁻ for neutral C₂O₂), confirming that in reality the dimer carries $-1e$ from the surface. Here you calculate the neutral system; the charge effect is introduced analytically via the CHE (see Section 3).

Coupling energy

$$\Delta G^0 = E(\text{slab} + \text{C}_2\text{O}_2) - E(\text{slab} + \text{CO}) - E(\text{CO}^{\text{gas}})$$

This is the free energy of the neutral coupling step *CO + CO(g) → *C₂O₂. After ZPE and entropy corrections it should be slightly negative (exothermic for the neutral system).

Post-relaxation check

After GEO_OPT, verify:

- d(C–C) ≈ 1.46 Å (single bond, paper value)
- O₁ still close to surface (bridge contact preserved)
- No dissociation of the dimer (d(C–C) < 1.8 Å)

3 (extra, not mandatory) Connecting Neutral DFT to Electrochemistry

Your three neutral calculations give ΔG^0 for the coupling step at $q=0$. To connect to the electrochemical scale you need the electron affinity (EA) of the dimer and the CHE potential term.

The key relation

The electrochemical coupling step is:



Its free energy at electrode potential U is:

$$\Delta G(U) = \Delta G^0 - \text{EA} - eU$$

Term	Meaning	How to obtain
ΔG^0	Coupling energy, neutral system	Your Group 3 DFT calculation
EA	Electron affinity of $^*\text{C}_2\text{O}_2$ on surface	From SI of paper: $\Delta H \approx 0.19$ eV for $\text{CO} + \text{CO}^- \rightarrow \text{C}_2\text{O}_2^-$
eU	Electrochemical potential	Set $U = 0$ V or $U = -0.40$ V (onset)

The onset potential is the value of U that makes $\Delta G(U) = 0$:

$$U_{\text{onset}} = (\Delta G^0 - \text{EA}) / e$$

The paper reports $U_{\text{onset}} = -0.40$ V. If your ΔG^0 is slightly different due to missing ZPE/solvation corrections, the onset will shift accordingly. The important point is the physical interpretation, not an exact match.

Results table to fill in

Quantity	Your DFT result	Paper reference
$\Delta E_{\text{ads}}(^*\text{CO})$ [raw DFT]	_____	-0.25 eV (with corrections)
$\Delta E_{\text{ads}}(2^*\text{CO})$ [raw DFT]	_____	-0.40 eV (with corrections)
ΔG^0 coupling [raw DFT]	_____	+0.81 eV ($2^*\text{CO} \rightarrow ^*\text{C}_2\text{O}_2$, with corr.)
U_{onset} [from $\Delta G^0 - \text{EA}$]	_____	-0.40 V
$d(\text{C}-\text{C})$ after relaxation	_____	1.46 Å

Phase 1.5 — ReaxFF Adsorption Energies on Large Slabs

Before running the computationally expensive metadynamics simulation, it is instructive to reproduce the three adsorption energies of Phase 1 using the ReaxFF force field on larger slabs. This serves two purposes: (1) it validates that the force field gives physically reasonable adsorption energies before committing to a long metadynamics run; (2) it shows students how the results scale with slab size, a convergence test that is too expensive to perform with DFT.

All Python scripts for this phase are pre-loaded on the MMM-Cloud AiiDALab instance prepared for this course. Log in, open the provided notebook, and run the cells in order. No local installation is required.

What to Compute

Repeat the three adsorption energy calculations of Phase 1 using LAMMPS + ReaxFF on slabs of increasing size, and compare with the DFT values obtained in Phase 1.

#	System	DFT value (Phase 1)	ReaxFF result
R1	$\Delta E_{\text{ads}}(*\text{CO})$ — slab sizes 3×3, 4×4, 5×5	? eV	_____
R2	$\Delta E_{\text{ads}}(2*\text{CO})$ — slab sizes 3×3, 4×4, 5×5	? eV	_____
R3	ΔG° coupling ($*\text{C}_2\text{O}_2$) — slab sizes 3×3, 4×4, 5×5	from Group 3 DFT	_____

Workflow on MMM-Cloud AiiDALab

The AiiDALab instance has three pre-loaded notebooks, one per calculation group. Each notebook contains: (1) the ASE slab-builder cell, (2) the LAMMPS energy-minimisation cell using the ReaxFF potential, and (3) the adsorption energy post-processing cell that prints the result and fills in the comparison table above. Run them in order:

- Log in to the MMM-Cloud AiiDALab instance provided by your instructor.
- Open the notebook `aiidalab-installations.ipynb` from the course folder, and run it.
- Open the notebook `aiidalab-reaxff-slabs.ipynb` from the course folder.
- Run all cells. Change over the three slab sizes (3×3, 4×4, 5×5) and print the adsorption energies for $*\text{CO}$, $2*\text{CO}$, and $*\text{C}_2\text{O}_2$ for each size.
- Fill the ReaxFF column in the results table above and compare with your DFT values from Phase 1.

Convergence check: if the adsorption energy changes by less than 0.05 eV between the 4×4 and 5×5 slabs, the result is considered converged with respect to lateral cell size. Use the converged slab size for the metadynamics in Phase 2.

Discussion Questions — Phase 1.5

- How do the ReaxFF adsorption energies compare with the DFT values? Are the trends ($\Delta E(*\text{CO})$ vs. $\Delta E(2*\text{CO})$ vs. ΔG° coupling) qualitatively reproduced?

13. How does the adsorption energy of *CO change with slab size? At what size do you consider the result converged, and what does this imply for the metadynamics slab choice?
14. ReaxFF energy minimisation is orders of magnitude faster than DFT. Estimate the speedup factor for the 5×5 slab and comment on when it would and would not be appropriate to use ReaxFF instead of DFT for this type of calculation.

Phase 2 — ReaxFF Metadynamics with LAMMPS + PLUMED

Collective Variable

CV = $d(\text{C}_1\text{--C}_2)$ — distance between the two carbon atoms

This CV directly tracks the C–C bond formation identified as the rate-determining step. The reactant state is $d \approx 3.5\text{--}4.0$ Å (two isolated *CO); the product state is $d \approx 1.46$ Å (*C₂O₂). The 1D FES along $d(\text{C--C})$ captures the essential physics while keeping the simulation tractable.

ReaxFF has no explicit electrons, so the metadynamics barrier corresponds to the neutral thermal coupling. The comparison with ΔG° from Phase 1 is therefore direct and consistent.

System Preparation

Build a 4×4×4 Cu(100) slab (64 atoms) with two CO molecules on adjacent top sites separated by $2 \times (a/\sqrt{2})$ along [110] (initial $d(\text{C--C}) \approx 7$ Å). Equilibrate at 300 K for 50 ps (NVT, Nosé–Hoover) before starting metadynamics.

LAMMPS + PLUMED Input

LAMMPS (key sections)

```
pair_style      reaxff NULL
pair_coeff      * * Cu_C_O_H.reax Cu C O
fix            NVT all nvt temp 300.0 300.0 100.0
fix            plumed all plumed plumedfile plumed.dat outfile plumed.out
```

PLUMED input (plumed.dat)

```
# C1 and C2: atom indices of the two C atoms (from your xyz file)
cc_dist: DISTANCE ATOMS=C1,C2

metad: METAD ARG=cc_dist ...
  PACE=500          # Gaussian every 500 steps
  HEIGHT=0.05       # initial height (eV)
  SIGMA=0.08        # Gaussian width (Å)
  BIASFACTOR=15     # well-tempered  $\gamma$ 
  TEMP=300.0
  GRID_MIN=1.2  GRID_MAX=4.5  GRID_BIN=330
... METAD

PRINT ARG=cc_dist,metad.bias STRIDE=100 FILE=COLVAR
```

FES Reconstruction and Results

```
plumed sum_hills --hills HILLS --outfile fes.dat --mintozero
```

Quantity	DFT ΔG° (Phase 1)	MetaD FES (Phase 2)
C–C distance, product well	1.46 Å (relaxed)	_____
Coupling ΔF (product – reactant)	ΔG° from Group 3	_____
Barrier $\Delta F^\ddagger$	N/A (no TS search)	_____

The metadynamics $\Delta F(\text{product} - \text{reactant})$ should be close to your ΔG° from Group 3 — this is the internal consistency check of the exercise. A large discrepancy signals either a force-field issue or incomplete metadynamics convergence.

5. Discussion Questions

15. Your ΔG° from Group 3 is the neutral coupling energy. How does including the electron (EA term) and the electrode potential (eU) change the picture? At what potential does the reaction become spontaneous?
16. The DFT calculations use PBE without D3 dispersion for the gas-phase CO reference, but with D3 for the slab. What systematic error does this introduce in ΔE_{ad} , and how could you correct it?
17. Compare $\Delta E_{\text{ad}}(*\text{CO})$ and $\Delta E_{\text{ad}}(2*\text{CO})$. What does the difference tell you about lateral interactions between adsorbed CO molecules? Is this consistent with the paper's discussion of coverage effects?
18. The metadynamics CV is $d(\text{C}-\text{C})$. Can you think of a case where this CV would fail to distinguish reactants from products on this surface?
19. ReaxFF has no explicit electrons. What physical effect is therefore missing in the metadynamics barrier compared to the DFT electrochemical barrier? How large do you expect this effect to be, based on your EA estimate?

6. Deliverables

Item	Description	Format
D1	CP2K input files for Groups 1, 2, and 3	Archive (.zip)
D2	Completed results table (Section 3)	PDF or Word
D3	FES plot with annotated barrier and wells	Figure + caption
D4	PLUMED input and COLVAR output	Plain text
D5	Written answers to all discussion questions	PDF or Word

7. Key References

1. Calle-Vallejo, F.; Koper, M. T. M. *Angew. Chem. Int. Ed.* 2013, 52, 7282–7285.
2. Peterson, A. A. et al. *Energy Environ. Sci.* 2010, 3, 1311–1315. (CHE formalism)
3. Nørskov, J. K. et al. *J. Phys. Chem. B* 2004, 108, 17886–17892. (CHE original)
4. Barducci, A.; Bussi, G.; Parrinello, M. *Phys. Rev. Lett.* 2008, 100, 020603. (well-tempered MetaD)
5. The PLUMED Consortium. *Nat. Methods* 2019, 16, 670–673.